MAM · Discrete Random Variables and the Binomial Distribution — Patterns and Solution Approaches

Sources: 27 past discrete random variable exam questions from 2016–2025, marking keys + exam reports (also cites the 2016 official sample paper `2016S-F4`; original paper in `papers/2016Sample_*`). Original questions in `topics/05_Discrete_RV.pdf`. Numerical examples in this document have been verified with Python. Review status: has undergone one round of independent review and revision according to feedback (see `Review_log.md`).

Marking-key reminders

- Recognise the binomial distribution and state the parameters: "State the distribution" usually awards 1 mark for the name and 1 mark for the parameters (occasionally combined as 1 mark, e.g. 2023A-Q13(b)); write it as X ~ Bin(5, 0.05). 2020 report: many did not recognise that 2020F-Q1 was a binomial distribution and made part (c), finding the mean and variance, overly complicated.
- Write the probability expression (e.g. P(X ≥ 4) = 1 − P(X ≤ 3)); most questions award 1 mark separately for this, so do not write only the calculator result.
- Keep intermediate probabilities to at least 4 decimal places (2020 report: keeping only 2 decimal places makes later answers inaccurate). For very small probabilities, keep enough significant figures: the answer to 2019A-Q18(c) is about 0.00003, and the key requires at least 5 decimal places.
- Distinguish between probability distribution and cumulative distribution: do not mix up tables of P(X = x) and P(X ≤ x) (2023 report).
- For "Explain / justify / state assumptions" questions, answer in context (2020 report specifically named 9(c) and 12(h)).

Pattern 1: Completing a distribution table / finding unknown probabilities

Approach

1. Equally likely cases: list the sample space, e.g. use a 4×4 table for the sum of two four-sided dice (2016A-Q15); for sampling without replacement use the multiplication rule or combinations.
2. When there are unknowns a, b: solve simultaneously using $\Sigma P = 1$ (1 mark) and $E(X) = \Sigma x \cdot P = \mu$ (1 mark) (2019A-Q10).
3. Construct a distribution table from raw data: frequency ÷ total number (2018F-Q4).
4. Given the cumulative distribution P(X ≤ x): P(X = k) = P(X ≤ k) − P(X ≤ k−1) (2024F-Q3).
5. New variable (e.g. number of unsold cakes Y = max(0, 3 − X), i.e. Y = 3 − X when X ≤ 3 and Y = 0 when X ≥ 3): Y = 0 corresponds to X ≥ 3 ⇒ P(Y=0) = P(X=3) + P(X=4) (2020A-Q9(c)).

Past questions: 2016A-Q15, 2018F-Q1(a), 2018F-Q4, 2019A-Q10, 2020A-Q9, 2024F-Q3, 2025A-Q8(a).

Pattern 2: Expectation, variance and linear transformations

Formulas: $E(X) = \Sigma x \cdot p$; $Var(X) = \Sigma x^2 p - [E(X)]^2$; $E(aX + b) = aE(X) + b$; $Var(aX + b) = a^2 Var(X)$; $SD(aX + b) = |a| \cdot SD(X)$.

Common traps

- Var(25 − 2X) = 4Var(X): the coefficient is $(-2)^2 = 4$, and the constant 25 has no effect on the variance (key wording: uses a positive factor of four).
- Scaling to a new mean and standard deviation (2016A-Q17(c)): use the standard deviation first to determine a ($|a| \cdot 22 = 15 \Rightarrow a = 15/22$; take $a > 0$ to preserve the rank order), then use the mean to determine b ($75a + b = 60 \Rightarrow b = 195/22$). The first mark in the key is "determines change on standard deviation first".
- Costs rise by 20% and then a $1 surcharge is added (2018A-Q12(f)) ⇒ new mean = 1.2μ + 1, new standard deviation = 1.2σ (adding a constant does not affect the standard deviation).

Past questions: 2016A-Q17, 2017A-Q19(c), 2018A-Q12(d)–(f), 2019A-Q10(b), 2023F-Q3(d), 2025A-Q8(b).

Pattern 3: Expected profit from games / fundraising ★ Frequently tested (examined in 6 of the last 10 years: 2017, 2020, 2021, 2022, 2024, 2025)

Recognition: "expected profit/loss", "who is better off in the long term", "what should be charged", "is it a good idea for the organisers".

Standard process

1. Define the return variable (from whose perspective: the player or the organiser).
2. Make a table of "net return for each outcome + probability".
3. E(return) = Σvalue × probability.
4. Draw a conclusion: E < 0 means the player loses in the long run; say "on average / in the long run".
5. Pricing: set expected profit = target (e.g. 20% of the charge) and solve; "minimum charge so that expected profit is not negative" = expected payout (2025A-Q8(c): E(X) = 0.8, $5 per red ball ⇒ charge at least $4).

Mark-earning points (2017A-Q13, 9 marks in total): (a)–(c): ✓ top row of the return table ✓ probabilities ✓ expected value expression ✓ expected value ✓ new expected payout ✓ conclusion (who is better off) ✓ explain the meaning of expectation; (d): ✓ set up equation ✓ solve for the charge.

Past questions: 2017A-Q13, 2020A-Q9(b), 2021A-Q13(c), 2022A-Q9(c)–(f), 2024A-Q14(c)–(e), 2025A-Q8(c).

Pattern 4: Bernoulli and the binomial distribution

Recognition: fixed number of trials n, only two outcomes per trial, trials independent, success probability p constant, and the question asks for "number of ...".

Approach

1. Write $X \sim Bin(n, p)$, and state what n and p mean in the context.
2. $P(X = k) = C(n,k)p^k(1-p)^{n-k}$; CF papers require by-hand calculation (2018F-Q1(e): Bin(5, 0.8), $P(X=2) = 10 \times 0.8^2 \times 0.2^3 = 32/625$).
3. $E(X) = np$, $Var(X) = np(1-p)$. Work backwards from the mean and variance to find n and p (2024F-Q4(b)): $np = ½$, $np(1-p) = 5/12 \Rightarrow 1 - p = 5/6 \Rightarrow p = 1/6$, $n = 3 \Rightarrow P(W=1) = 3 \times (1/6) \times (5/6)^2 = 25/72$.
4. Get the boundaries for "at least / at most / more than" correct: more than three ⇒ $X \geq 4$; system fails if fewer than 2 work ⇒ 4 or more break down.
5. Change the parameters when the context changes: after one alarm is removed it becomes Bin(4, 0.05) (2019A-Q18(d)).

Assumptions question (2 marks): Prefer these two — the objects are mutually independent; each object's success/failure probability is the same and constant (the keys for 2019A-Q18(b) and 2020A-Q12(h) both give only these two categories). "Only two outcomes" and "n is fixed" are generally given in the question, so they should not be written as assumptions. Write them in context, e.g. "the failure of one alarm does not affect another". 2020A-Q12(h) also requires discussing whether the assumptions are reasonable. Why it is not binomial (2024A-Q14(b), 2025A-Q8(d)): the number of trials is not fixed; sampling without replacement means trials are not independent and p changes. Improvement method: sample with replacement.

Past questions: 2016S-F4, 2018F-Q1, 2019A-Q18, 2020F-Q1, 2021A-Q13, 2022A-Q9, 2025A-Q15.

Pattern 5: Combined with the normal distribution ("at least k out of n items ...")

Recognition: first use the normal distribution to find the probability p that one item meets the condition, then ask how many out of n items meet it.

Two-step method

1. Normal: $p = P(X > 3.7)$. Keep enough decimal places.

2. Binomial: Y ~ Bin(20, p), find P(Y ≥ 10). The key requires writing "binomial" and the parameters; usually 1 mark for the name and 1 mark for the parameters (2017A-Q19(b), 2018A-Q12(b), 2023A-Q12(b)).

Past questions: 2017A-Q19(b), 2018A-Q12(b), 2019A-Q11(b), 2021A-Q8(d), 2023A-Q12(b); 2023A-Q13(b) is similar, but the single-item probability comes from a uniform distribution.

Pattern 6: Path / combinatorial counting type (robot, Lucky 7)

Approach: First translate the event into a condition on the "number of successes". For example, the robot returns to the origin ⇔ number of forward steps = number of backward steps (with an odd number of steps it is impossible to return to the origin, so the probability is 0, and the reason must be written); forward ≥ 20 cm ⇔ number of forward steps − number of backward steps ≥ 4 ⇔ at least 7 forward steps out of 10, then calculate using the binomial distribution (P ≈ 0.3823).

Past questions: 2025A-Q15. Related counting or geometric probability: 2021A-Q10(b) (list the combinations with sum ≥ 7, weighted sum gives p = 7/24, without using the binomial distribution), 2023A-Q13(a) (θ has a uniform distribution, geometric probability α/π).